

Estimated Annual Agricultural Pesticide Use Pacific Northwest (WA/OR/ID), Regional Ranking 2014-2018

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Summary

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides across **Pacific Northwest (WA/OR/ID)** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data, summed across every county in Washington, Oregon, and Idaho. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, each compound’s chemical class (abbreviated; see the key in Table 1 and the full classification in `data/pesticide_classification.csv`, shared with each state’s own chapters), and — for comparison — its 2018 rank *within each state considered alone*. Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” [-Pesticide National Synthesis Project webpage](#)

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”); summing three states together inherits this caveat at an even larger scale.
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 included once a complete 2019 release is available.
- **Comparing raw mass across states is not apples to apples** — WA, OR, and ID farm very different total acreages and crop mixes, so a compound’s regional rank reflects total farmland as much as anything else. The per-state rank columns below are about *where a compound concentrates*, not a claim about which state uses more pesticide overall.

A compound’s regional rank and its individual state ranks can differ substantially — a compound used heavily across Idaho’s potato acreage but barely at all in Oregon will rank high regionally only if Idaho’s volume carries it; **NA** in a state’s rank column means the compound was not among that state’s own top 80 for 2018.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
ALS	ALS-Inhibitor (Sulfonylurea & Related)
AVM	Avermectin
BIO	Biological/Microbial
BIP	Bipyridylum
CB	Carbamate

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
CAA	Chloroacetanilide
DMI	DMI Fungicide (Triazole/Imidazole)
DIA	Diamide
DNA	Dinitroaniline
DTC	Dithiocarbamate
FUM	Fumigant
GLY	Glycine/Phosphonate
INO	Inorganic/Mineral
IGR	Insect Growth Regulator
NEO	Neonicotinoid
OC	Organochlorine
OP	Organophosphate
FUN	Other Fungicide
HRB	Other Herbicide
INS	Other Insecticide/Miticide
OTH	Other/Unclassified
PHX	Phenoxy
PHP	Phenylpyrazole
PGR	Plant Growth Regulator
PYR	Pyrethrin/Pyrethroid
SDH	SDHI
SPN	Spinosyn
QOI	Strobilurin (QoI)
TC	Thiocarbamate
TRZ	Triazine

Table 2: Top 80 EPest-high Estimates, Pacific Northwest (WA/OR/ID), 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) and Each State's Own Rank

Rank (PNW)	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg	Rank (WA)	Rank (OR)	Rank (ID)
1	METAM	FUM	17641358	17105484	15766434	1	1	2
2	METAM POTASSIUM	FUM	9919151	6214504	5370421	4	7	1
3	DICHLOROPROPENE	FUM	7814442	8173642	8385906	2	2	5
4	PETROLEUM OIL	INS	5927510	7202551	6524079	3	5	31
5	CHLOROPICRIN	FUM	4670658	1909295	1222471	6	3	3
6	GLYPHOSATE	GLY	2494939	2475098	2381611	7	6	6
7	SULFURIC ACID	INO	2475480	5192844	5473738	11	9	4
8	BACILLUS AMYLOLIQUEFACIEN	BIO	2414155	1352449	854675	8	4	7
9	SULFUR	INO	2200223	1876399	1811938	5	8	24
10	CALCIUM POLYSULFIDE	INO	1165336	1059340	928334	9	11	61
11	KAOLIN CLAY	INO	942600	1626351	1253148	10	25	93
12	2,4-D	PHX	693525	609818	594210	12	12	8
13	EPTC	TC	494327	530335	565282	14	21	9
14	PARAQUAT	BIP	423478	335664	284070	15	10	28
15	COPPER	INO	368382	278903	227944	13	20	92
16	MANCOZEB	DTC	366732	447175	534308	16	13	17
17	CHLOROTHALONIL	FUN	314079	335039	310397	17	22	11
18	PENDIMETHALIN	DNA	307838	327621	299146	24	24	10
19	MCPA	PHX	307476	336751	333934	19	17	13
20	BROMOXYNIL	HRB	244238	241215	263335	18	27	19
21	METRIBUZIN	TRZ	240846	226849	222835	26	33	12
22	METOLACHLOR & METOLACHLOR-S	CAA	239440	223407	241410	22	26	14
23	METOLACHLOR-S	CAA	219422	200875	210207	23	28	18
24	CHLORPYRIFOS	OP	193991	231947	237271	20	31	22
25	BURKHOLDERIA SPP	BIO	179237	156890	118128	25	16	32
26	DIMETHENAMID & DIMETHENAMID-P	CAA	173528	182968	155629	31	43	15
27	DIMETHENAMID-P	CAA	173528	151165	136548	32	44	16
28	ATRAZINE	TRZ	168123	229365	167913	30	19	25
29	GLUFOSINATE	GLY	163446	86955	75677	21	23	46
30	ETHOPROPHOS	OP	129193	292700	190837	47	14	NA
31	TRIFLURALIN	DNA	124501	104364	122152	59	15	29
32	PROPICONAZOLE	DMI	118595	117376	98097	28	37	44
33	METHOMYL	CB	104964	112804	100724	44	36	26
34	OXAMYL	CB	104655	48542	109821	27	39	50
35	FLUROXYPYR	HRB	98586	98368	92624	41	52	21
36	THIOPHANATE-METHYL	FUN	93678	117090	142895	34	41	43
37	DICAMBA	HRB	84414	87037	75191	48	35	33
38	COPPER HYDROXIDE	INO	83518	102668	92226	33	40	81
39	COPPER SULFATE	INO	82796	112690	102101	38	30	132
40	MCPB	PHX	80136	27975	18809	29	71	74

Table 2: Top 80 EPest-high Estimates, Pacific Northwest (WA/OR/ID), 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) and Each State's Own Rank (*continued*)

Rank (PNW)	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg	Rank (WA)	Rank (OR)	Rank (ID)
41	MALEIC HYDRAZIDE	PGR	78675	64027	73626	53	164	23
42	ACETOCHLOR	CAA	75429	37265	37002	NA	110	20
43	AZOXYSTROBIN	QOI	73373	66847	58924	60	34	36
44	DIURON	HRB	70007	147970	146415	61	29	47
45	DCPA	HRB	66878	107589	100887	45	38	54
46	DIQUAT	BIP	65262	47886	46276	46	51	35
47	BACILLUS THURINGIENSIS	BIO	61011	41460	32064	35	79	124
48	BENTAZONE	HRB	60959	58336	68015	39	60	59
49	DIMETHOATE	OP	60301	100003	107431	40	47	56
50	PHOSPHORIC ACID	INO	60258	82441	160529	63	NA	27
51	BOSCALID	SDH	55550	65573	68025	51	54	37
52	FLUFENACET	CAA	55418	43651	40381	175	18	NA
53	POTASSIUM BICARBONATE	INO	55213	71542	53224	36	115	179
54	PYRACLOSTROBIN	QOI	54823	61164	63140	52	48	41
55	CARBARYL	CB	53808	46490	67043	37	99	140
56	CLOPYRALID	HRB	52827	43819	44439	49	42	57
57	ETHALFLURALIN	DNA	49241	82853	82529	55	88	34
58	COPPER OXYCHLORIDE	INO	47644	35089	25864	42	46	90
59	PYRIMETHANIL	FUN	35836	33163	31560	137	75	30
60	LINURON	HRB	34697	30862	21970	43	180	103
61	CLETHODIM	HRB	34429	28252	25201	66	59	52
62	HEXAZINONE	TRZ	31592	43583	42171	88	107	39
63	ZIRAM	DTC	31569	51545	49363	192	32	NA
64	PROPARGITE	INS	31403	20278	41996	80	91	42
65	TRI-ALLATE	TC	30685	65673	59743	67	129	49
66	TRIFLUMIZOLE	DMI	30532	32235	32986	50	62	142
67	FLUAZINAM	FUN	30363	23031	25550	54	80	73
68	FLUTOLANIL	SDH	30179	28435	29582	56	65	69
69	MALATHION	OP	29675	36132	46031	69	45	80
70	SIMAZINE	TRZ	29182	16583	27004	58	49	NA
71	MEFENOXAM	FUN	29041	26749	23764	62	76	62
72	IMIDACLOPRID	NEO	28169	31886	40009	82	67	48
73	FLUOPYRAM	SDH	27322	22151	17862	78	63	53
74	PINOXADEN	HRB	26959	24070	22980	99	57	51
75	PENTHIOPYRAD	SDH	24773	25305	24453	57	78	119
76	HYDROGEN PEROXIDE	INO	24484	10606	17266	153	163	38
77	TRICLOPYR	HRB	24464	23884	23554	65	50	159
78	FLUXAPYROXAD	SDH	22396	22628	16964	72	83	60
79	QUINTOZENE	FUN	21783	25116	32727	NA	NA	40
80	METOLACHLOR	CAA	20019	22533	31203	NA	111	45

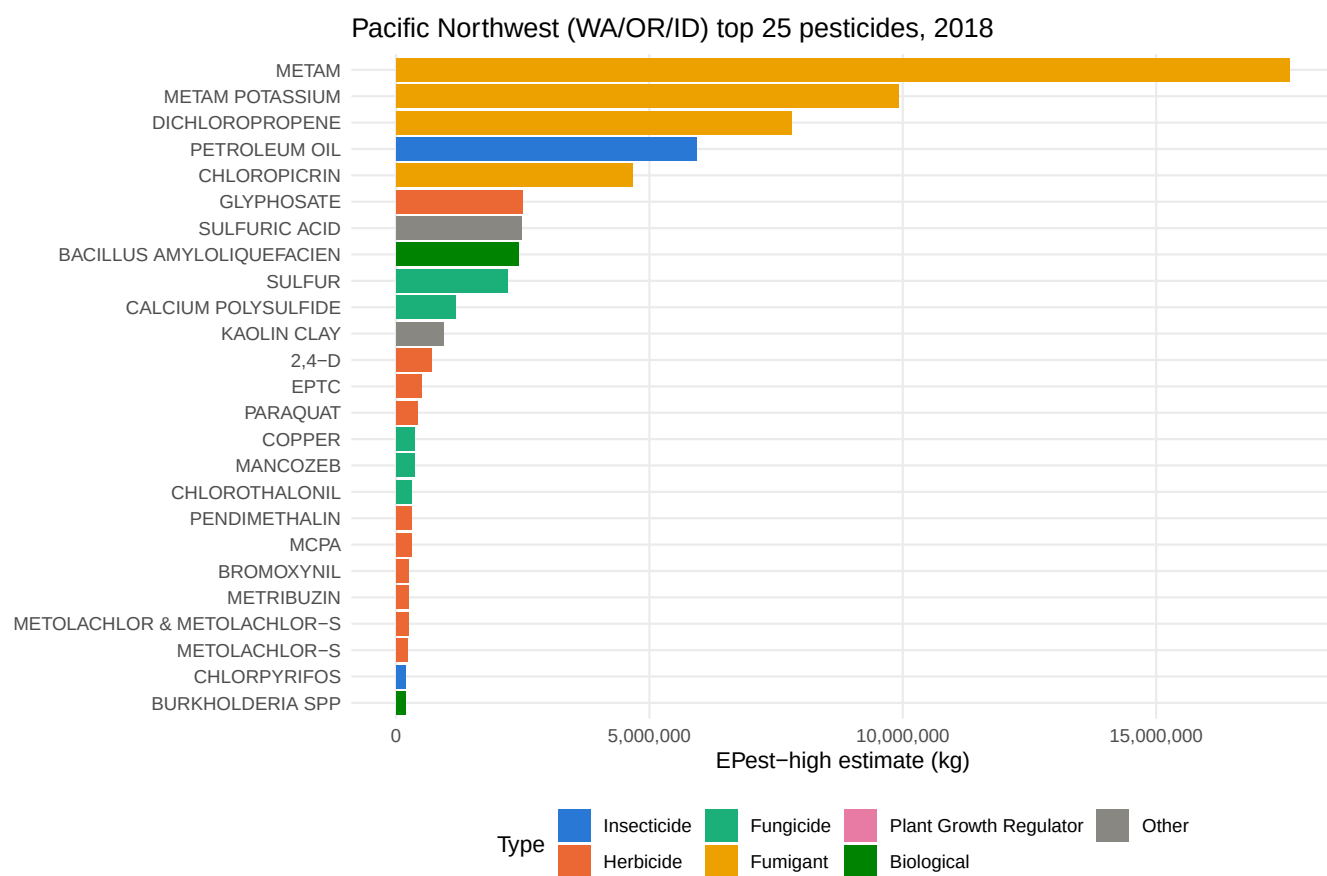


Figure 1: Top 25 compounds region-wide by Epest-high mass applied, most recent year, colored by pesticide type.

Appendices

Table 3: Top 50 Pacific Northwest (WA/OR/ID) Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM	14135510	17641358
2	METAM POTASSIUM	8618129	9919151
3	DICHLOROPROPENE	7352926	7814442
4	PETROLEUM OIL	5880875	5927510
5	CHLOROPICRIN	1134191	4670658
6	GLYPHOSATE	2289327	2494939
7	SULFURIC ACID	1463266	2475480
8	BACILLUS AMYLOLIQUEFACIEN	926523	2414155
9	SULFUR	2153654	2200223
10	CALCIUM POLYSULFIDE	1162214	1165336
11	KAOLIN CLAY	939335	942600
12	2,4-D	610826	693525
13	EPTC	316098	494327
14	PARAQUAT	293725	423478
15	COPPER	364185	368382
16	MANCOZEB	333546	366732
17	CHLOROTHALONIL	268201	314079
18	PENDIMETHALIN	231363	307838
19	MCPA	268289	307476
20	BROMOXYNIL	226538	244238
21	METRIBUZIN	189686	240846
22	METOLACHLOR & METOLACHLOR-S	152539	239440
23	METOLACHLOR-S	144878	219422
24	CHLORPYRIFOS	182921	193991
25	BURKHOLDERIA SPP	71335	179237
26	DIMETHENAMID & DIMETHENAMID-P	130494	173528
27	DIMETHENAMID-P	130494	173528
28	ATRAZINE	84306	168123
29	GLUFOSINATE	160450	163446
30	ETHOPROPHOS	36969	129193
31	TRIFLURALIN	8562	124501
32	PROPICONAZOLE	110157	118595
33	METHOMYL	55256	104964
34	OXAMYL	76815	104655
35	FLUROXYPYR	92170	98586
36	THIOPHANATE-METHYL	74568	93678
37	DICAMBA	51171	84414
38	COPPER HYDROXIDE	59501	83518
39	COPPER SULFATE	81022	82796
40	MCPB	10302	80136
41	MALEIC HYDRAZIDE	31776	78675
42	ACETOCHLOR	38912	75429
43	AZOXYSTROBIN	50427	73373
44	DIURON	36672	70007
45	DCPA	35493	66878
46	DIQUAT	53634	65262
47	BACILLUS THURINGIENSIS	60554	61011
48	BENTAZONE	50150	60959
49	DIMETHOATE	41312	60301
50	PHOSPHORIC ACID	43472	60258

Table 4: Top 50 Pacific Northwest (WA/OR/ID) Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	15292963	17105484
2	DICHLOROPROPENE	7667660	8173642
3	PETROLEUM OIL	6707796	7202551
4	METAM POTASSIUM	4935983	6214504
5	SULFURIC ACID	3127033	5192844
6	GLYPHOSATE	2337580	2475098
7	CHLOROPICRIN	564509	1909295
8	SULFUR	1782267	1876399
9	KAOLIN CLAY	1199314	1626351
10	BACILLUS AMYLOLIQUEFACIEN	590559	1352449
11	CALCIUM POLYSULFIDE	1027074	1059340
12	2,4-D	535059	609818
13	EPTC	359536	530335
14	MANCOZEB	431166	447175
15	MCPA	257413	336751
16	PARAQUAT	222964	335664
17	CHLOROTHALONIL	305926	335039
18	PENDIMETHALIN	249804	327621
19	PROPAMOCARB HCL	25475	308421
20	ETHOPROPHOS	49088	292700
21	COPPER	273490	278903
22	BROMOXYNIL	214027	241215
23	CHLORPYRIFOS	202850	231947
24	ATRAZINE	103287	229365
25	METRIBUZIN	194029	226849
26	METOLACHLOR & METOLACHLOR-S	131394	223407
27	METOLACHLOR-S	121510	200875
28	DIMETHENAMID & DIMETHENAMID-P	133027	182968
29	BURKHOLDERIA SPP	63134	156890
30	DIMETHENAMID-P	117441	151165
31	DIURON	40786	147970
32	PROPICONAZOLE	108164	117376
33	THIOPHANATE-METHYL	82856	117090
34	METHOMYL	71773	112804
35	COPPER SULFATE	104709	112690
36	DCPA	73534	107589
37	TRIFLURALIN	10176	104364
38	COPPER HYDROXIDE	78223	102668
39	DIMETHOATE	50872	100003
40	FLUROXYPYR	77509	98368
41	DIMETHENAMID	46759	95409
42	DICAMBA	55050	87037
43	GLUFOSINATE	77878	86955
44	ETHALFLURALIN	52190	82853
45	PHOSPHORIC ACID	57107	82441
46	POTASSIUM BICARBONATE	54608	71542
47	AZOXYSTROBIN	48518	66847
48	TRI-ALLATE	17263	65673
49	BOSCALID	55204	65573
50	MALEIC HYDRAZIDE	27988	64027

Table 5: Top 50 Pacific Northwest (WA/OR/ID) Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	14099549	15766434
2	DICHLOROPROPENE	7602245	8385906
3	PETROLEUM OIL	6221391	6524079
4	SULFURIC ACID	3371390	5473738
5	METAM POTASSIUM	4225158	5370421
6	GLYPHOSATE	2270128	2381611
7	SULFUR	1707111	1811938
8	KAOLIN CLAY	988328	1253148
9	CHLOROPICRIN	387502	1222471
10	CALCIUM POLYSULFIDE	901395	928334
11	BACILLUS AMYLOLIQUEFACIEN	381833	854675
12	2,4-D	518598	594210
13	EPTC	400844	565282
14	MANCOZEB	520876	534308
15	MCPA	274636	333934
16	CHLOROTHALONIL	289328	310397
17	PROPAMOCARB HCL	25475	308421
18	PENDIMETHALIN	240540	299146
19	PARAQUAT	180126	284070
20	BROMOXYNIL	220235	263335
21	METOLACHLOR & METOLACHLOR-S	137923	241410
22	CHLORPYRIFOS	207837	237271
23	COPPER	219820	227944
24	METRIBUZIN	189326	222835
25	METOLACHLOR-S	123884	210207
26	ETHOPROPHOS	38353	190837
27	ATRAZINE	81890	167913
28	PHOSPHORIC ACID	86681	160529
29	DIMETHENAMID & DIMETHENAMID-P	114153	155629
30	DIURON	47422	146415
31	THIOPHANATE-METHYL	94819	142895
32	DIMETHENAMID-P	104801	136548
33	TRIFLURALIN	9740	122152
34	BURKHOLDERIA SPP	47570	118128
35	OXAMYL	96503	109821
36	DIMETHOATE	55203	107431
37	COPPER SULFATE	89168	102101
38	DCPA	63627	100887
39	METHOMYL	65788	100724
40	PROPICONAZOLE	90133	98097
41	DIMETHENAMID	46759	95409
42	FLUROXYPYR	75962	92624
43	COPPER HYDROXIDE	74251	92226
44	ETHALFLURALIN	61167	82529
45	GLUFOSINATE	55562	75677
46	DICAMBA	48628	75191
47	MALEIC HYDRAZIDE	32788	73626
48	BOSCALID	58702	68025
49	BENTAZONE	61178	68015
50	CAPTAN	23220	67231